PROGRAM NOTE

Neil Thapen's "Pink Trombone" is a popular web teaching tool that allows users to manipulate a simplified vocal tract: by moving the tongue, glottis, nasal passages, and lips with your mouse, you can make recognizable — if uncanny — phonemes, the building blocks of language out. I discovered Zack Qattan's excellent set of tools that allows text-to-speech with the tool, and adapted it to create a piece for "pink trombone choir." The text is adapted from T.S. Eliot's "Burnt Norton", the first of his *Four Quartets* (1936).This piece was written for the Peabody Laptop Ensemble.

WHAT THE PIECE NEEDS

- **Eight players, at least.** Two on each of the four parts. More is better and nothing needs rewriting: the parts share their written pitches out among whoever turns up, so extra players fill out a chord rather than doubling a line. Sixteen is comfortable. One player per part works, but sounds thinner than intended.
- **A laptop for each player.** Any operating system, any current browser. No downloads or extra software necessary.
- **One more laptop for the conductor.** It runs the small server the others connect to and shows the cue desk. It makes no sound of its own. The conductor does not need to be positioned on stage – all cues are served to each player's browser.
- **Amplification the players can hear.** This is not a headphone piece. Most of what the score asks for is players listening to each other and matching — speed, density, pitch. If they cannot hear the room, the piece does not happen. A house PA, or powered speakers spread across the ensemble.
- **A network everyone is on.** Wired or wireless, house or a travel router of your own. It does not need to reach the internet, but if a stable public network exists already it can be used.**This is not recommended - create a local LAN if possible!**
- **A projector, optionally.** It can be driven from the conductor's laptop.

GETTING EVERYONE CONNECTED

1. On the conductor's laptop, open a Terminal and navigate to the Tract folder. Run the command: `node scan/scan-bridge.js`. It gives itself a certificate for whatever addresses this machine has today, so the address it prints starts with `https://` - which is what lets the players' laptops run the fast audio path.
2. It prints the addresses to hand out. Players get the bare one — `192.168.1.42:8080` or whatever it says — and type that and nothing else.
3. The conductor opens the same address with `/conduct.html` on the end. The projector laptop uses `/stage.html`.
4. macOS asks whether to allow incoming connections the first time. Allow it.

5. Each player types their name and chooses their part from the panel that appears. The laptop remembers both, so it is asked once and not again — the name is what labels that player on the desk. A laptop that is always the same part can skip that step with `?part=2` on the end of the address.
6. **Every player clicks once anywhere on their page.** No browser will make a sound until the page has been clicked. Until it is, everything looks correct and nothing is audible. Choosing a part when opening the page should serve this function.
7. Volume up, and check the sound is leaving by the output you think it is.
8. On the desk, the roster along the top says how many machines are on each part. A part reading *0 · nobody* is a laptop that never arrived. Sort it out before you start rather than during.

The strips along the bottom of the desk are the ensemble, one per laptop, labelled with the name each player typed when they joined. Each shows the words of the phrase that player is on and where their playhead has got to inside them, lit while they are sounding — pink for voiced, green for whispering. Four parts at once is more texture than you can take apart by ear from the front of a room; this is how you see who is where, who is still on the last word, and who has whisper on when they should not.

Before the first downbeat

The written pitches are shared out among whoever is connected, and that settles at the first quiet moment. **Get everyone on before you begin**, and try not to add a laptop in the middle of a phrase — it re-deals the pitches underneath players who are already sounding.

Someone who reloads, or arrives late, is handled without you doing anything: the desk repeats the standing cue quietly every few seconds, so they pick up what the ensemble is doing with no false downbeat.

WHO DOES WHAT

ROLE	HOW MANY	
Conductor	one	Runs the piece and the server.
Players	two or more per part	Four parts. No one needs to read music.
Projection	optional, one	A laptop on `/stage.html`, plugged into the projector. It can run on the Conductor computer.

The parts differ by register and by what they are asked to do, not by difficulty.

CONDUCTING

PREP, then DOWNBEAT. Both are the space bar.

Players can already see what is coming: the next cue's text appears underneath what they are doing the moment the last downbeat lands, so there is nothing to read in a hurry. PREP turns the named players' lights amber — *soon* — and DOWNBEAT flashes the light white and swaps the new instruction in. `esc` takes back a prep nobody has acted on. The pointer moves on by itself after each downbeat, so conducting a run is space, space, space.

Moving the pointer with `←` `→` changes what the players see as coming next, so skipping a cue in rehearsal is visible to them at once. The cue you last actually sent stays on the desk under *last sent*, whatever the pointer has moved on to.

A cue speaks only to the parts written under it. A part not named in a cue is sent nothing at all — no instruction, no light, no flash. That is deliberate, and it is what lets some passages be conducted while the players around them are listening to each other and deciding for themselves. *Carry on with what you are doing* and *there is no cue for you here* have to be different things.

The timing is yours. Nothing in this piece is metered and nothing counts. Cue when the texture is ready, and let a passage last as long as it is still going somewhere. Two performances need not be the same length.

THE SCREEN IS THE SCORE

Every player reads their current instruction off their own screen, cued live. These printed pages are what the screen cannot tell you: the setup, the shape, the way in. The phrase-by-phrase score at the back is printed from the same file the players are reading, so the two cannot drift apart — but if you ever find them disagreeing, **the screen is right**.

A WAY INTO REHEARSAL

Not in order. The piece opens with its most exposed music and gets easier to hold together as it goes, which is the wrong direction to learn it in.

1. **Phrase 3 first.** Everybody does the same thing at the same time. It teaches sweeping with the mouse, and it teaches the ensemble to watch each other's speed, with nothing else going on.
2. **Then phrase 1.** Staggered entries, held keys instead of the mouse. It teaches scanning, and it teaches players to hear their own entry inside a texture instead of waiting to be told.
3. **Then phrase 4.** Short, everyone together, whisper on and then off — a good place to make the whisper toggle an ensemble gesture rather than a private one.

4. **Then the dense ones, 2 and 7**, where nobody synchronises and the instruction is a density rather than a shape. Rehearse these by listening to the whole and adjusting, not by getting your own part right.
5. **Phrase 5 last.** It asks for free pitch inside a band with everyone moving independently, and it is the only passage where a player can be genuinely lost without noticing.

TWO-MINUTE SOUNDCHECK

First, walk the room and read the pill in the top corner of every player's page. It should say `bridge: … · worklet`. If it says `· main thread` in amber, that laptop is running the synth the slow way — it will still play, but it is working much harder than it needs to. Almost always it means that player typed `http://` instead of `https://`.

Then the sound itself. Everyone on phrase 1. One part at a time — I, then II, then III, then IV — each holding A alone for a few seconds while the rest stay silent. You are listening for four clearly separated registers at roughly even loudness. It catches a muted output, a laptop on the wrong part and a machine that never woke up, in one pass.

WHERE PLAYERS GET LOST

- **Whisper left on** from the previous phrase. Z latches and nothing resets it.
- **Scanned off the end** of the phrase and waiting for something to happen. Press the key again.
- **Wandered out of the band** in a drifting phrase. The lit keys on the piano are the whole of what is allowed.
- **Still holding the spacebar** after the cue to let go.

The cue sheet in these pages is printed from `cues.json` and `parts.json` — the files the players read live. Change a cue and run `node score/make-score.js` again rather than editing this document.